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Finding practical
solutions to real life
problems with
software [*Dave Farley*]

What makes a
good developer?

Understanding
Ask questions
Courage
Curiosity
Respect

Decompose the problem
What are the steps
Where to start
What is the challenge
Small steps



Code
Execution context
OS
CPU
Storage
Network
Physical server



Real-world environment
User idea, mindset
Requirements
Analysis
Architecture
Algorithms
Code

Become the machine
Computer as a self-
contained world where the
programmer is a ruler.



Write once,
run anywhere

Simple C
OOP
Type level programming
Functional programming
Backend, a.k.a SpringBoot
TDD
Beyond Java

Simple C

Write C-like programs
without worrying about
memory

Object Oriented Programming

Reduce complexity by abstraction

Bundle data with behaviors and communication

Model the world in a context of your code

Type-level programming

Make you code describe itself

Use compiler to verify correctness

Functional programming

Program as transformation

No state changes

Performance loss

Backend a.k.a. SpringBoot

Build "serious" distributed applications

Communicate over network

Use databases

Speak JSON to outside world

Test Driven Development

Write test before the code

Make the code testable

Build safety net against breaking
changes

Automate everything

Beyond Java

Java Virtual Machine

Other JVM languages

Scala, Kotlin, Clojure, Groovy